Threatening Patch Attacks on Object Detection in Optical Remote Sensing Images

Xuxiang Sun, Gong Cheng, *Member, IEEE*, Lei Pei, Hongda Li, and Junwei Han, *Fellow, IEEE*

***Abstract*— Advanced patch attacks (PAs) on object detection in natural images have pointed out the great safety vulnerability in methods based on deep neural networks (DNNs). However, little attention has been paid to this topic in optical remote sensing images (O-RSIs). To this end, we focus on this research, i.e., PAs on object detection in O-RSIs, and propose a more threatening patch attack (TPA) without the scarification of the visual quality. Specifically, to address the problem of inconsistency between the local and global landscapes in existing patch selection schemes, we propose leveraging the first-order difference (FOD) of the objective function before and after masking to select the subpatches to be attacked. Furthermore, considering the problem of gradient inundation when applying existing coordinate-based loss (CBL) to PAs directly, we design an IoU-based objective function specific for PAs, dubbed bounding box (Bbox) drifting loss (BDL), which pushes the detected Bboxes far from the initial ones until there are no intersections between them. Finally, on two widely used benchmarks, i.e., DIOR and DOTA, comprehensive evaluations of our TPA with four typical detectors (faster R-CNN, FCOS, RetinaNet, and YOLO-v4) witness its remarkable effectiveness. To the best of our knowledge, this is the first attempt to study the PAs on object detection in O-RSIs, and we hope this work can get our readers interested in studying this topic.**

***Index Terms*— Adversarial patch attacks (PAs), object detection, remote sensing images.**

I. Introduction

DRAWING on the powerful representation ability of deep neural networks (DNNs), a great deal of revolutionary achievements have been made in the aspect of image understanding technology [1], [2], [3], [4], [5], [6], [7], [8], [9]. Similarly, the technology of understanding optical remote sensing images (O-RSIs) has made great progress [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20]. Nevertheless, the exposed adversarial vulnerability [21], [22] of the DNN leaves great security concerns, hindering their widespread applications.

Manuscript received 12 February 2023; revised 4 April 2023 and 21 April 2023; accepted 2 May 2023. Date of publication 5 May 2023; date of current version 22 May 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62136007, in part by the Natural Science Basic Research Program of Shaanxi under Grant 2021JC-16 and Grant 2023-JC-ZD-36, in part by the Fundamental Research Funds for the Central Universities, in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2021B1515020072, and in part by the Innovation Foundation for Doctor Dissertation of Northwestern Polytechnical University under Grant CX2022054. *(Corresponding author: Gong Cheng.)*

The authors are with the School of Automation, Northwestern Polytechnical University, Xi'an 710129, China, and also with the Research and Development Institute of Northwestern Polytechnical University in Shenzhen, Shenzhen 518057, China (e-mail: gcheng@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3273287



Fig. 1. **Visualization of samples drawn from DIOR and DOTA.** Here, we can see that the distribution of objects in remote sensing images is globally sparse and locally dense.

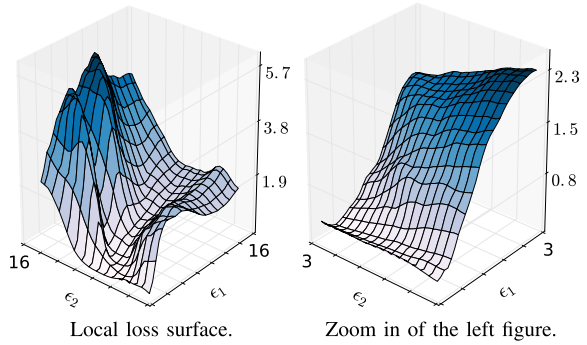
Facing this security hazard, many researchers devote themselves to studying adversarial robustness [21], [22], [23], [24], [25], [26]. Recently, the security concerns of DNN-based deep learning methods in O-RSIs have also received progressive attention [27], [28], [29], [30]. Among them, little attention has been paid to the adversarial vulnerability in O-RSI object detection, an essential and typical research field in O-RSI understanding.

To date, there are three widely studied attack schemes for object detection, i.e., the full-scale attacks (FSAs) [31], [32], [33], [34], [35], the patch attacks (PAs) [36], [37], [38], and the adversarial patches (APs) [39], [40], [41]. In short, FSAs perturb the whole image, and only the pixels in some specific regions are perturbed in the PAs. Compared with FSAs and PAs, the adversarial examples generated by APs are more human-perceptible, and all the targets in an image share the same pattern. The only difference among these APs is their physical parameters (location, angle, scale, and so on). Given the intrinsic property of O-RSIs, i.e., the objects in O-RSIs are globally sparse and locally dense (cf., Fig. 1 for better visualization), PAs exhibit more threats than the other attacks for O-RSIs, on account of their region efficiency and visual imperceptibility.

In general, the attack scheme of PAs consists of two critical steps, i.e., the patch selection scheme and the objective function. For the former, recent research [38] proposes to leverage the norm of the gradients passed from the objective function to select the most critical subpatches. However, adding perturbation is to produce a significant function drop, but the gradient is defined within a tiny neighborhood of data points. Thus, it could not simulate the masking manipulation in the

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:42:26 UTC from IEEE Xplore. Restrictions apply.



Local loss surface. Zoom in of the left figure.

Fig. 2. **Illustrations of the local curvature artifacts [26].** It reflects the loss of the adversarially trained Inception-v3 [1] ($v3_{adv}$), with the form of $\boldsymbol{x}^* = \boldsymbol{x} + \epsilon_1 \cdot \boldsymbol{g} + \epsilon_2 \cdot \boldsymbol{g}^{\perp}$. Here, $\boldsymbol{g}$ is the signed gradient of $v3_{adv}$, and $\boldsymbol{g}^{\perp}$ is an orthogonal adversarial direction, sampled from inception-v4 [2].

PAs. Besides, previous study [26] has shown that the direction of gradients does not always align with the optimal direction (cf., Fig. 2 for more details), that is, for high-dimensional nonlinear functions such as DNNs, the magnitude of the gradient at a point is not always positively correlated with the variation degree of the loss function globally. To this light, we propose a patch selection scheme based on the first-order difference (FOD), which calculates the FOD of the objective function by masking the subpatches and selects top-k subpatches with the largest FOD. In this way, we can obtain the variation degree of the loss function within a larger neighborhood by covering the subpatches and use this metric as a measure of whether the region is favorable to attack.

Besides, the objective functions leveraged in existing PAs [36], [37], [38] only focus on the classification branch without the attack on the bounding box (Bbox) regression. Fortunately, a commonly adopted loss [33] in the field of FSAs designs the coordinate-based loss (CBL) to make all the Bboxes out of the image. However, the perturbed regions in the PAs are not large enough to force the detected Bboxes out of the image, especially for the images with the characteristic of global sparse such as O-RSIs (cf., Fig. 1). Then, the gradient of CBL will exist over the entire attack progress with a larger magnitude than that passed from the classification branch [33]. Consequently, the gradients passed from the classification head may be at risk of being inundated by those passed from the regression branch. To this end, we propose a Bbox drifting loss (BDL) to merely reduce the intersection over union (IoU) between the detected Bboxes and the initial ones so as to avoid the problem of gradient inundation when applying CBL to PAs directly.

Finally, we validate the effectiveness of our method on DIOR [10] and DOTA [11], respectively. Here, a total of seven typical detectors are utilized to evaluate the general effectiveness of our method. Specifically, four kinds of victim detectors, including faster R-CNN [3], RetinaNet [5], FCOS [4], and Yolo-v4 [6], are leveraged for the evaluations, and we equip the first three detectors with two backbones, i.e., ResNet-50 [8] with feature pyramid networks (FPNs) [42] and ResNet-101 [8] with FPNs [42]. Throughout the comprehensive evaluations, our TPA achieves the most threatening results.

In summary, our contributions are as follows.

1) The threats of the PAs on object detection in O-RSIs are exhibited for the first time in this article, which provides the preliminary empirical evidence for the safety concern when applying DNN-based methods to practical deployment.
2) We propose an FOD patch selection scheme to boost the visual efficiency of the PAs. It imitates the attack scheme in the PAs by masking the subpatches and selecting the ones with the highest FOD.
3) We propose BDL, an IoU-based objective function specialized for PAs. In this way, the gradient passed from the regression branch can be stopped if there are no overlaps between the detected Bboxes and the initial ones.

II. Related Work

In this section, we first review some classical adversarial attacks for image recognition. Later, we will pay much attention to the studies on adversarial attacks for object detection.

A. Adversarial Attacks on Image Recognition

Early studies on adversarial vulnerability mainly focus on the task of image recognition. Among them, the white-box setting [21], [22], [23], [24], [25], where the attackers have full access to the victim model, received a widespread attention. Later, on the black-box setting, some researchers tried to enhance the transferability via data augmentation [43], [44], [45], advanced optimization scheme [45], [46], and so on. Besides, some other attack scenarios also received in-depth studies, e.g., query-based black-box attacks [47], [48] and model stealing [49], [50]. There is a wide variety of works on this topic. Limited by the space, we could not review them thoroughly in this article. To this end, we recommend our readers [51] for more comprehensive reviews of the research progress.

B. Adversarial Attacks on Object Detection

1) Full-Scale Attacks: Xie et al. [31] propose a first attempt, dubbed DAG, to attack object detection and segmentation. It sets a wrong category for the target to increase the confidence of negative samples while reducing the confidence of positive samples in an iterative manner. Inspired by Carlini and Wagner [23] and the expectation over transformation (EoT) [52], Chen et al. [32] propose to add random perturbation over each iteration so as to enhance the robustness of adversarial examples for faster R-CNN [3]. Later, Li et al. [33] design robust adversarial perturbation (RAP) to attack the region proposal networks (RPNs) in two-stage object detectors by destroying both the classification and regression. In addition to the objective function of RAP [33], Zhang et al. [34] introduce contextual loss to increase the confidence of the background and inhibit the confidence of the foreground. Recently, Nezami et al. [35] propose to precisely manipulate the pixel of the target object to change its label without affecting the other objects.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:42:26 UTC from IEEE Xplore. Restrictions apply.

2) Patch Attacks: In this field, Wu et al. [36] propose a diffused patch with the shapes like asteroid or grid and pay more attention to the proposals that escaped from attack. Zhao et al. [37] design heatmap-based and consensus-based algorithms to select patches for the attack. Recently, Huang et al. [38] propose RPAttack to enhance the threats of PAs in the patch selection and optimization schemes. Specifically, it proposes a patch selection based on the gradient feedback (GF) and leverages the ensemble learning to improve the attack strength.

3) Adversarial Patches: Brown et al. [39] are the first attempt to attack object detection via a single AP. It designs an unrestricted patch with a fixed position to attack the classification branch. Liu et al. [40] mislead the victim detector by forcing it to perceive only the stoked rectangular patch at a fixed position. Liu et al. [53] proposed a perceptual-sensitive generative adversarial network to synthesize APs. Moreover, there are a great variety of studies on APs, including aerial detection [54], [55], [56] and physical APs for real-world detection [41], [57], [58].

In summary, current works regarding adversarial attacks on object detection pay much attention to FSAs and APs, whereas the research on PAs has not received widespread attention. However, given the threats posed by the imperceptibility of PAs, it deserves in-depth research. Therefore, we take a closer look at PAs on object detection in this article.

III. Preliminary

A. Problem Formulation

Considering the general representation, we denote the ground truth of an image $\boldsymbol{x}$ as follows:

$$\boldsymbol{O}(\boldsymbol{x}) = \left\{\boldsymbol{B}_i(\boldsymbol{x}), \boldsymbol{y}_i(\boldsymbol{x})\right\}, \quad (i = 1, 2, 3, \ldots, N) \tag{1}$$

where N denotes the number of instances in $\boldsymbol{x}$. Here, $\boldsymbol{B}_i(\boldsymbol{x}) = \{B_i^x, B_i^y, B_i^w, B_i^h\}$ is the location information, with (B_i^x, B_i^y) denoting the coordinates of the Bbox center point, and (B_i^w, B_i^h) are the width and height of the Bbox. The class information $\boldsymbol{y}_i(\boldsymbol{x}) \in \{1, 2, \ldots, C\}$ denotes the label of an instance, where C is the number of categories. In this way, the detected results for a given image $\boldsymbol{x}$ can be represented as follows:

$$\widetilde{\boldsymbol{O}}(\boldsymbol{x}) = \left\{\widetilde{\boldsymbol{B}}_i(\boldsymbol{x}), \widetilde{\boldsymbol{P}}_i(\boldsymbol{x})\right\} \tag{2}$$

where $\widetilde{\boldsymbol{P}}_i(\boldsymbol{x}) = \{\widetilde{P}_i^1, \widetilde{P}_i^2, \ldots, \widetilde{P}_i^C\}$ is the class probability vector. Then, the detected class is the index corresponding to the maximum in $\widetilde{\boldsymbol{P}}_i(\boldsymbol{x})$, that is,

$$\widetilde{C}_i(\boldsymbol{x}) = \text{argmax } \widetilde{\boldsymbol{P}}_i(\boldsymbol{x}). \tag{3}$$

Based on the above notations, the objective of adversarial attacks on object detection can be formulated as the following:

$$\begin{aligned} &\min \ ||\boldsymbol{\xi}||_p \\ &\text{s.t. } \boldsymbol{O}(\boldsymbol{x}_{\text{adv}}) \neq \boldsymbol{O}(\boldsymbol{x}) \end{aligned} \tag{4}$$

where $\boldsymbol{x}$ are clean images and $\boldsymbol{\xi}$ are the corresponding perturbations. The objective in (4) is to find the adversarial example with the minimum visual distortion. Since both our method and our competitors are based on the BIM optimizing framework, we choose the ℓ_∞-norm as the visual constraint, i.e., $p = \infty$ in (4).

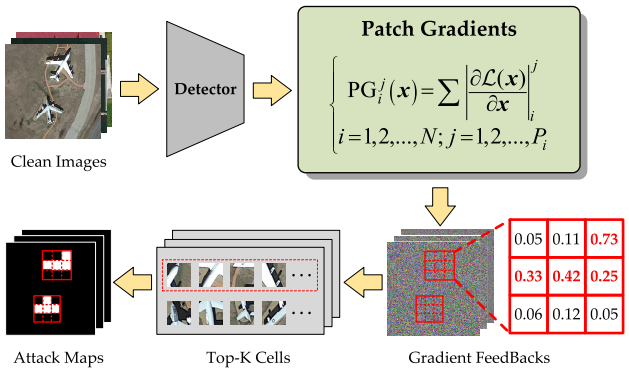


Fig. 3. Illustration of the patch selection scheme in RPAttack [38].

Here, the misleading to either the branch of classification or regression can be seen as a successful attack, that is, either $\widetilde{C}_i(\boldsymbol{x}) \neq \widetilde{C}_i(\boldsymbol{x}_{\text{adv}})$ or $\text{IoU}(\widetilde{\boldsymbol{B}}_i(\boldsymbol{x}), \widetilde{\boldsymbol{B}}_i(\boldsymbol{x}_{\text{adv}})) < 0.5$ can be seen as a successful attack. Besides, for PAs, the representation of adversarial examples can be formulated as follows:

$$\boldsymbol{x}_{\text{adv}} = \boldsymbol{x} + \mathbf{M}^* \odot \boldsymbol{\xi}^* \tag{5}$$

where $\odot$ is the elementwise multiplication. $\mathbf{M}$ is the attack map with the same size as $\boldsymbol{x}$, which determine the regions to be attacked. Here, $\boldsymbol{\xi}^*$ represents the optimal solution of (4), and $\mathbf{M}^*$ denotes the optimal results of the attack map, which is fixed over the entire attack progress in general.

B. Patch Selection Scheme

In this section, we mainly introduce the patch selection scheme in RPAttack [38]. For simplicity, we dub it GF. As can be seen in Fig. 3, it first splits the Bbox of each instance evenly to get a series of subpatches. Then, the subpatches are ranked with respect to the gradient norm, and the top-k subpatches with the largest gradient norms are selected. Here, $\mathcal{L}$ denotes the objective function leveraged for the optimization of the perturbations, and $\mathrm{PG}_i^j(\boldsymbol{x})$ denotes the ℓ_1-norm of the jth subpatch in the ith instance of an image $\boldsymbol{x}$.

C. Objective Function

Here, the objective function proposed by [33] to attack the regression branch of a detector is introduced in this section. For simplicity, we dub it the CBL. In general, attacking the regression branch aims at making the detected Bboxes own undesirable shape or position. To this end, RAP assigns large offsets for the detected Bboxes to make them out of the image as much as possible. Formally, the CBL is expressed as follows:

$$\begin{aligned} \mathcal{L}_{\text{CBL}}(\widetilde{\boldsymbol{B}}(\boldsymbol{x}_{\text{adv}})) = \sum_{j=1}^{m} z_j \big(&\left(\widetilde{B}_j^x - \tau^x\right)^2 + \left(\widetilde{B}_j^y - \tau^y\right)^2 \\ &+ \left(\widetilde{B}_j^w - \tau^w\right)^2 + \left(\widetilde{B}_j^h - \tau^h\right)^2\big) \end{aligned} \tag{6}$$

where $\{\tau^x, \tau^y, \tau^w, \tau^h\}$ is the predefined offsets. Besides, z_j is the indicator of the jth proposal, and its formulation can be

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:42:26 UTC from IEEE Xplore. Restrictions apply.

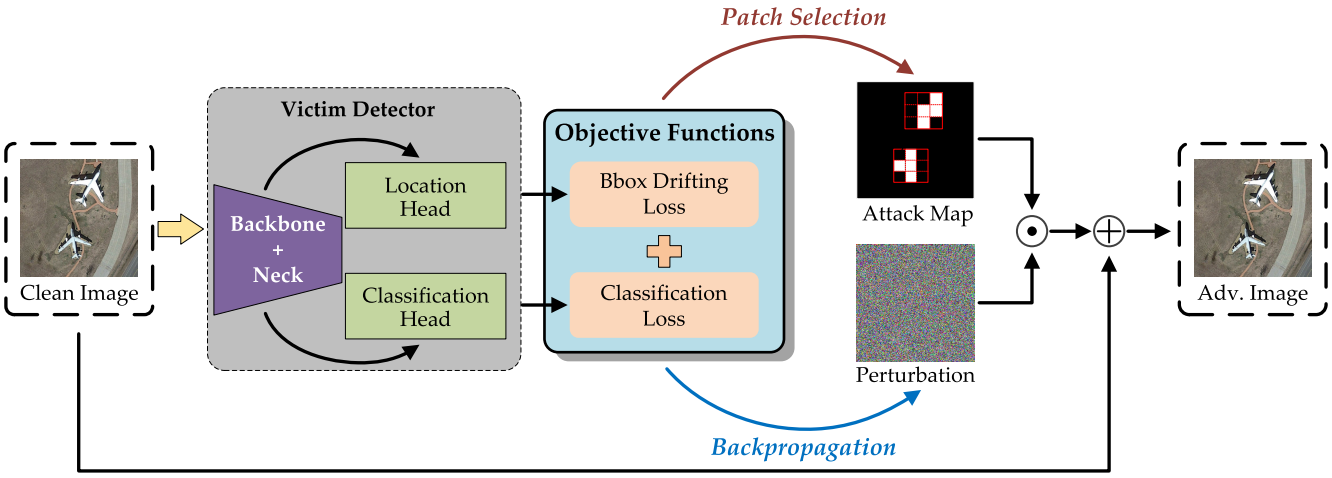


Fig. 4. **Framework of our proposed TPA.** Here, $\odot$ and $\oplus$ denote the elementwise multiplication and addition, respectively. In addition, the generated adversarial examples should be clipped to a legal interval of the image, as shown in (5). Please refer to Section IV-A for a more exact illustration.

expressed as follows:

$$\begin{cases} z_j = 1, & \text{if } \begin{cases} \text{IoU}(\widetilde{\boldsymbol{B}}_j(\boldsymbol{x}), \widetilde{\boldsymbol{B}}_j(\boldsymbol{x}_{\text{adv}}))0.1 \\ \max\ \widetilde{\boldsymbol{P}}_j(\boldsymbol{x}_{\text{adv}}) > 0.4 \end{cases} \\ z_j = 0, & \text{otherwise.} \end{cases} \tag{7}$$

In this article, we follow the setting of RAP [33] to set $\tau^x = \tau^y = \tau^w = \tau^h = 10^5$ in (7).

IV. Method

A. Overview

This article aims to propose a threatening PA that is applicable to multiple target detectors, including faster R-CNN [3], RetinaNet [5], YOLO-v4 [6], and FCOS [4]. Since the structures of these detector networks are different, the attack framework is not designed specifically for a certain kind of detector but is based on the final prediction results. Here, we depict the framework of our TPA in Fig. 4. Specifically, we first select the regions to acquire the attack map $\mathbf{M}$ via the proposed FOD patch selection scheme, and $\mathbf{M}$ will be fixed over the entire attack progress. Then, similar to BIM [59], we optimize the adversarial example in an iterative manner.

Formally, at each iteration $t+1$, we have

$$\begin{aligned} \boldsymbol{x}_{\text{adv}} &= \hat{\boldsymbol{x}}_{t+1} \\ &= \boldsymbol{x} + \mathbf{M} \odot \boldsymbol{\xi}_{t+1} \end{aligned} \tag{8}$$

where $\boldsymbol{\xi}_{t+1}$ is formulated as follows:

$$\boldsymbol{\xi}_{t+1} = \text{Clip}_{-\epsilon}^{\epsilon}\left\{\text{Clip}_0^1\{\hat{\boldsymbol{x}}_t + \mathbf{M} \odot [\alpha \cdot \text{sign}(\boldsymbol{g}_{t+1})]\} - \boldsymbol{x}\right\} \tag{9}$$

where ϵ denotes the ℓ_∞ constraint, $\text{sign}(\cdot)$ is the sign function, and α is the attack step size. In our TPA, the attack map $\mathbf{M}$ is a binary mask, which determines where to attack. Besides, the expression of $\boldsymbol{g}_{t+1}$ in (9) is

$$\boldsymbol{g}_{t+1} = \frac{\nabla_{\hat{\boldsymbol{x}}_t}\mathcal{L}(\hat{\boldsymbol{x}}_t)}{||\nabla_{\hat{\boldsymbol{x}}_t}\mathcal{L}(\hat{\boldsymbol{x}}_t)||_1}. \tag{10}$$

Here, the specific details regarding how to get the attack map $\mathbf{M}$ and the expression of $\mathcal{L}(\boldsymbol{x})$ will be introduced next.

B. FOD Patch Selection Scheme

For PAs, the selection of subpatches is a critical factor that affects the attack efficiency. However, as we have mentioned in Section I, the advanced patch selection scheme proposed by RPAttack [38] may suffer from the problem of inconsistency between the local and global landscapes, since the gradient is defined within a small neighborhood around the data point, which is not large enough to explore the global landscapes. To this end, we propose to imitate the "masking" manipulation in PAs by covering each subpatch of the instance and select the subpatches with the highest feedback.

Specifically, as shown in Fig. 5, we first divide the Bbox of each instance into a grid of $n \times n$. Here, considering the objects in O-RSIs with the same class could have different sizes, n can vary according to the size of the instance. In this article, we provide two options for the grid segmentation, i.e., the uniform segmentation scheme and the scale-adaptive segmentation scheme. All the instances in the uniform segmentation share the same setting of n. For simplicity, we denote $\text{U}(n)$ as the uniform segmentation scheme with the size of n. For the scale-adaptive segmentation scheme, similar to MS COCO [60], we divide the instances into three kinds of scales in terms of their areas. To be specific, we denote the area of an instance as $\mathcal{S}$. Then, we set n to n_1 for $\mathcal{S} \leq 32^2$, n_2 for $32^2 < \mathcal{S} \leq 64^2$, and n_3 for $\mathcal{S} > 64^2$. Thus, we can use $\text{SA}(n_1, n_2, n_3)$ to represent the scale-adaptive segmentation scheme with certain parameters. The results regarding these choices will be reported and analyzed in Section V-C.

Once we acquire the masked inputs, we feed them into the victim detector to calculate the FOD. Formally, for a subpatch, we define the FOD as follows:

$$\begin{aligned} \text{FOD}_i^j(\boldsymbol{x}) &= \mathcal{L}(\boldsymbol{x}) - \mathcal{L}\left(\boldsymbol{x}_i^j\right) \\ &= \mathcal{L}(\boldsymbol{x}) - \mathcal{L}\left(\mathbf{M}_i^j \odot \boldsymbol{x}\right) \end{aligned} \tag{11}$$

where $\mathbf{M}_i^j$ denotes masking the jth subpatch of the ith instance. Correspondingly, $\boldsymbol{x}_i^j$ is the input with the jth subpatch of the ith instance that is masked, and $\text{FOD}_i^j(\boldsymbol{x})$ is the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:42:26 UTC from IEEE Xplore. Restrictions apply.

FOD of the jth subpatch belonging to the ith instance. Associating with the objective function utilized in our TPA (please refer to Section IV-C for a more detailed introduction), (11) can be expressed as follows:

$$\left(\max \widetilde{\boldsymbol{P}}_i(\boldsymbol{x}) - \widetilde{\boldsymbol{P}}_i^{\widetilde{C}_i(\boldsymbol{x})}\left(\boldsymbol{x}_i^j\right)\right) + \left(1 - \mathrm{IoU}\left(\widetilde{\boldsymbol{B}}_i(\boldsymbol{x}), \widetilde{\boldsymbol{B}}_i\left(\boldsymbol{x}_i^j\right)\right)\right) \tag{12}$$

where $\widetilde{\boldsymbol{P}}_i^{\widetilde{C}_i(\boldsymbol{x})}(\boldsymbol{x}_i^j)$ denotes the $\widetilde{C}_i(\boldsymbol{x})$th entry of $\widetilde{\boldsymbol{P}}_i(\boldsymbol{x}_i^j)$.

C. Bounding Box Drifting Loss

Compared with the image classifier, attacking the object detector is deemed as a more complicated problem, on account of the complex outputs from the detector. In addition to attacking the classification branch, destroying the regression branch of the detector is another essential step, which could enhance the threats of attacks to a large extent. Existing methods for PAs only focus on the attack against the classification. Fortunately, RAP [33] has proposed CBL to break the Bbox regression branch. However, as we analyzed in Section I, applying CBL directly to the framework of PAs may be at risk of gradient inundation. In other words, since PAs only attack some specific regions of an image, making all the detected Bboxes out of the entire image seems to set an *impossible* objective. As a result, the gradients of CBL will exist over the entire attack progress. Besides, the norms of the gradient passed from CBL are larger than that passed from the classification loss, on account of the large threshold in CBL. Consequently, the gradient of the classification loss may suffer from being inundated by that of CBL, leading to the stagnation of optimization.

In fact, with the goal of attacking the regression branch, offsetting the detected Bboxes is exactly what we want to see. To this end, there exist many solutions for this purpose. One of the most threatening scenarios is that there are no intersections between the initial Bbox and the detected ones after attacking. Thus, we formulate the attack on the regression branch as the above situation, that is, drifting the detected Bboxes away from the initial one until there is no overlap between them. When it comes to measuring the overlaps between two Bboxes, a natural idea is to leverage the IoU, a commonly adopted metric in object detection. Therefore, we formulate our BDL as follows:

$$\mathcal{L}_{\mathrm{BDL}}(\hat{\boldsymbol{x}}) = \frac{1}{N}\sum_{i=1}^{N} \max(\mathrm{IoU}(\widetilde{\boldsymbol{B}}_i(\boldsymbol{x}), \widetilde{\boldsymbol{B}}_i(\hat{\boldsymbol{x}}))) \tag{13}$$

where N denotes the number of instances in the initial results. During the iterations, there may exist a lot of detected Bboxes around the initial one. To this end, (13) takes the Bbox with the highest IoU between the initial one into calculation. In this way, the detected Bboxes that have no overlaps between the initial one are not taken into consideration, that is, these Bboxes have been attacked successfully.

Another loss function utilized in our TPA is for the attack on the classification branch. Here, we use the loss in RPAttack [38], which is formulated as follows:

$$\mathcal{L}_{\mathrm{cls}}(\hat{\boldsymbol{x}}) = \frac{1}{k}\sum_{i=1}^{k} |\max \widetilde{\boldsymbol{P}}_i(\hat{\boldsymbol{x}})|^2 \tag{14}$$

where k denotes the number of detected results in $\hat{\boldsymbol{x}}$. Finally, we use (15) as the total objective function in our TPA

$$\mathcal{L}(\hat{\boldsymbol{x}}) = \mathcal{L}_{\mathrm{BDL}}(\hat{\boldsymbol{x}}) + \mathcal{L}_{\mathrm{cls}}(\hat{\boldsymbol{x}}). \tag{15}$$

TABLE I
RESULTS ON DIOR-A AND DOTA-A DATASETS

Detector	FR-50	FR-101	FC-50	FC-101	RT-50	RT-101	YOLO-v4
mAP (%)	88.30	88.60	87.30	87.60	87.30	87.30	89.50
Recall (%)	90.30	90.90	91.30	91.60	92.80	92.80	90.00
mAP (%)	68.70	68.40	65.70	66.80	62.20	64.80	69.70
Recall (%)	77.70	76.10	79.10	80.00	79.50	81.30	76.80

Here, the results above the double lines are those on DIOR-A dataset and the others are those on DOTA-A dataset, the same to Tab. II

V. EXPERIMENTS

A. Experimental Settings

1) Datasets: We carry out the evaluations on two widely adopted benchmarks for object detection in O-RSIs, i.e., DIOR [10] and DOTA [11]. The DIOR dataset covers 192 518 instances of 20 categories, which are formatted in a fixed size of 800 × 800. The DOTA dataset (we use DOTA-1.0 in this article) includes 2860 images with the size varying across 800 × 800 to 4000 × 4000, covering 15 categories. In practical, we split the images into a fixed size, i.e., 800 × 800. Besides, to facilitate the evaluation of the following research regarding adversarial attacks on object detection in O-RSIs, similar to the commonly adopted protocol [38], [46] in the field of adversarial attacks, we sample 2000 images from the testing subset of DIOR and the validation subset of DOTA, respectively, dubbed DIOR-A and DOTA-A. The classwise instance distributions of them are exhibited in Fig. 6. Here, we plot both the classwise instance distribution histograms and their corresponding kernel density estimation (KDE) curves, in which we can see that the sampled datasets share almost the same classwise distribution with their corresponding parent datasets. Besides, since only 2000 images are utilized for the evaluation, they could reduce the calculation budget to some extent, compared with using the parent datasets for the evaluation. Thus, we will carry out all the following experiments on these sampled datasets.

2) General Settings: We leverage the PyTorch framework to implement our method on a single NVIDIA RTX 2080Ti. Here, four kinds of detectors are utilized for the evaluations, where faster R-CNN [3] (FR), RetinaNet [5] (RT), and FCOS [4] (FC) are trained on MMDetection, and YOLO-v4 [6] is based on DarkNet [7]. For the detectors on MMDetection, we equip them with two backbones, including ResNet-50 [8] with FPN [42] and ResNet-101 [8] with FPN [42]. Here, FR-50 denotes faster R-CNN with ResNet-50 + FPN as the backbone and so on. The testing results on clean images of DIOR-A and DOTA-A are reported in Table I.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:42:26 UTC from IEEE Xplore. Restrictions apply.

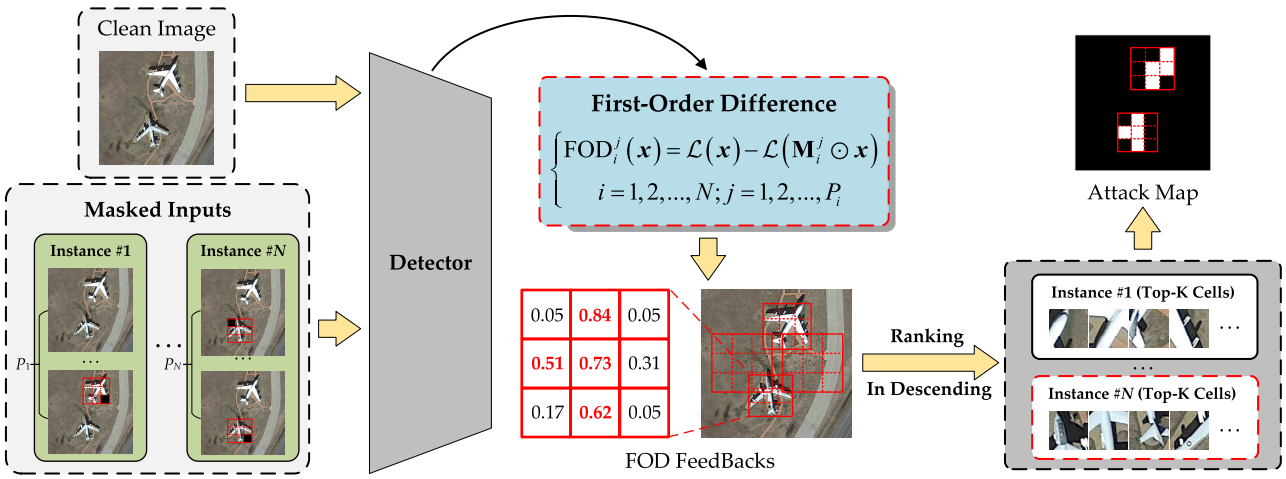


Fig. 5. Illustration of the proposed FOD patch selection scheme.

TABLE II

COMPARISON RESULTS ON DIOR-A AND DOTA-A DATASETS. RESULTS ARE SEPARATED BY THE DOUBLE LINE, ABOVE WHICH ARE THE RESULTS ON DIOR-A AND THE REMAINING ARE ON DOTA-A. THE BEST RESULTS ARE SHOWN IN **BOLD**

Detector	FR-50		FR-101		FC-50		FC-101		RT-50		RT-101		YOLO-v4	
Method	RPA [38]	Ours	RPA [38]	Ours	RPA [38]	Ours	RPA [38]	Ours	RPA [38]	Ours	RPA [38]	Ours	RPA [38]	Ours
mAP (%)	53.20	**47.80**	55.80	**50.80**	-	**47.00**	-	**51.90**	-	**39.60**	-	**44.70**	50.40	**20.30**
Recall (%)	69.70	**55.80**	71.00	**58.30**	-	**61.50**	-	**66.40**	-	**58.00**	-	**60.70**	29.60	**20.20**
ℓ_0 Norm	0.110	**0.059**	0.120	**0.059**	-	**0.059**	-	**0.059**	-	**0.059**	-	**0.059**	0.110	**0.059**
ℓ_2 Norm	6.840	**4.990**	6.860	**5.160**	-	**5.640**	-	**5.790**	-	**5.170**	-	**5.310**	6.840	**4.860**
mAP (%)	31.90	**26.60**	54.00	**39.40**	-	**28.70**	-	**32.10**	-	**17.10**	-	**20.50**	34.80	**21.70**
Recall (%)	49.80	**26.60**	65.50	**48.20**	-	**47.60**	-	**50.30**	-	**37.00**	-	**40.50**	36.40	**22.30**
ℓ_0 Norm	0.060	**0.032**	0.060	**0.031**	-	**0.033**	-	**0.032**	-	**0.032**	-	**0.032**	0.060	**0.032**
ℓ_2 Norm	5.412	**3.628**	5.557	**3.756**	-	**4.304**	-	**4.411**	-	**3.944**	-	**4.005**	5.412	**3.731**

Since RPAttack [38] (RPA) utilizes Faster R-CNN [3] and YOLO-v4 [6] to carry out an ensemble attack, it can not attack RetinaNet and FCOS under the white-box setting. In this case, we only report the results of RPAttack on FR-50, FR-101, and YOLO-v4.

Here, for DIOR-A, we utilize the train–val subsets to train the victim detectors, and the training subset is leveraged to train the detectors for DOTA-A. More details and the sampled datasets are open accessed.[1]

3) Attack Settings: Considering the implementation and relativity, we choose RPAttack [38], the state-of-the-art PA on object detection in natural images, as our competitor. For the evaluation metrics, we leverage mAP and Recall to measure the strength of different attacks. Besides, we introduce median ℓ_2- and ℓ_0-norms to evaluate the visual quality of the adversarial examples. For the attack settings, the ℓ_∞ constraint ϵ in (9) is set to $10/255$, and the number of iterations T is 10. The step size α in (9) is set to $1/255$. Furthermore, both RPAttack and our TPA attack $\lfloor (n \times n)/2 \rfloor$ patches of an instance, where we divide each instance into $n \times n$ patches in total. Here, $\lfloor \cdot \rfloor$ represents the floor division. Finally, for the grid segmentation, we use SA(1, 2, 3) as the main scheme. The reason for this choice will be discussed in Section V-C.

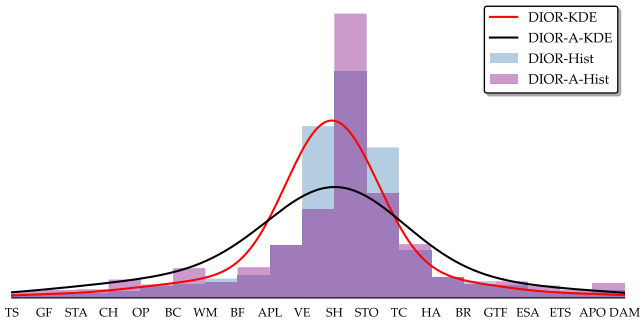
B. Peer Comparisons

In this section, we carry out experiments to validate the advancement of our TPA. The quantitative results are summarized in Table II. Surprisingly, the performance of our TPA can surpass the advanced competitor, RPAttack, which leverages the ensemble setting to enhance its threats, where ensemble setting is a powerful attack setting that utilizes more than one victim for the optimization of adversarial examples. Specifically, we summarize the advantages of our TPA as threefolds. First, we exceed RPAttack by a large margin in terms of Recall, which indicates that TPA owns the great potential of hiding targets than RPAttack. Second, we can keep the highest attack strength while keeping the lowest ℓ_2-norms, which demonstrates the great attack efficiency of our TPA. Finally, different from RPAttack that is restricted by the choice of the victim detector, the proposed TPA is applicable to a variety of detection models.

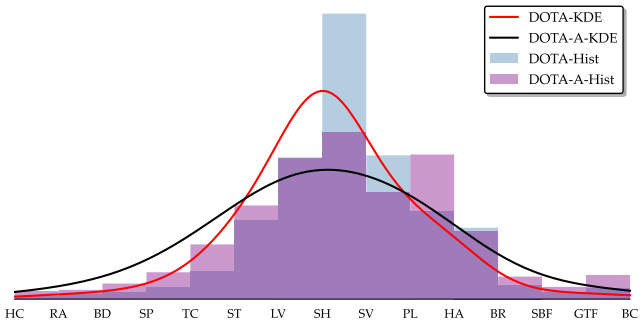
Later, we also visualize the detected results in Fig. 7. Here, all the targets surrounded by the boxes with the same color belong to the same category, and we use different colors to distinct the categories. As we can see from Fig. 7, compared with RPAttack, TPA can make most targets "invisible." In addition, Fig. 7 releases a significant trend that the invisible ability of RPAttack gets decreased as the density of the targets increases. From this point of view, TPA shows more threats than RPAttack, since the dense instance distribution is a very common phenomenon in O-RSIs.

[1]https://github.com/plpl2019/TPA

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:42:26 UTC from IEEE Xplore. Restrictions apply.



DIOR v.s. DIOR-A w.r.t. Class-Wise Instance Distribution.



DOTA v.s. DOTA-A w.r.t. Class-Wise Instance Distribution.

Fig. 6. Classwise instance distributions.

TABLE III
ABLATION RESULTS REGARDING THE PATCH SELECTION SCHEME

Detector	Method	mAP (%)	Recall (%)	ℓ_0 Norm	ℓ_2 Norm
FR-50	RD	52.80	61.30	**0.059**	5.050
	GF [38]	50.00	59.40	**0.059**	4.990
	FOD	**47.80**	**55.80**	**0.059**	**4.900**
FC-50	RD	52.30	67.50	**0.059**	5.640
	GF [38]	49.60	65.00	**0.059**	5.620
	FOD	**47.00**	**61.50**	**0.059**	**5.510**
RT-50	RD	45.20	64.90	**0.059**	5.180
	GF [38]	43.10	62.90	**0.059**	5.180
	FOD	**39.60**	**58.00**	**0.059**	**5.170**

C. Further Studies

In this section, we arrange extended experiments to take a closer look at our TPA. Specifically, we provide three ablation studies regarding the choice of the patch selection scheme, the choice of the objective functions for the Bbox regression, and the choice of the patch segmentation scheme.

1) Ablation Study on Patch Selection Scheme: As we have introduced in Section I, RPAttack utilizes the GF to select the most critical regions to be attacked, which may suffer from the inconsistency between the local and global landscapes, leaving the attack efficiency to be suppressed. To this end, we vary the choice of the patch selection scheme in our TPA to see the effect of our FOD. The results of this part are summarized in Table III. Here, GF stands for the GF patch selection scheme that is leveraged in RPAttack [38], and RD represents selecting the regions in a random manner. Not surprisingly, FOD achieves the best results in terms of both strength and visual quality, and picking the subpatches in a random manner exhibits the poorest performance, that is, selecting the subpatches indeed plays a critical role in final results. Besides, as we have pointed out in Section I, FOD selects subpatches via imitating the attack scheme in PAs to escape from the suboptimal within the local neighborhood. From this point of view, a relatively constrictive neighborhood could mislead the choice of subpatches, resulting in poor threats.

TABLE IV
ABLATION RESULTS REGARDING THE OBJECTIVE FUNCTION FOR THE BBOX REGRESSION

Detector	Method	mAP (%)	Recall (%)	ℓ_0 Norm	ℓ_2 Norm
FR-50	$\mathcal{L}_{\text{cls}}$	55.20	61.70	**0.059**	**4.800**
	$\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{CBL}}$	50.90	61.50	**0.059**	5.340
	$\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{BDL}}$	**47.80**	**55.80**	**0.059**	4.990
FC-50	$\mathcal{L}_{\text{cls}}$	55.80	70.06	**0.059**	**5.466**
	$\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{CBL}}$	52.60	65.30	**0.059**	5.630
	$\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{BDL}}$	**47.00**	**61.50**	**0.059**	5.540
RT-50	$\mathcal{L}_{\text{cls}}$	42.90	61.90	**0.059**	**5.010**
	$\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{CBL}}$	42.60	61.80	**0.059**	5.240
	$\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{BDL}}$	**39.60**	**58.00**	**0.059**	5.170

2) Ablation Study on Bbox Regression Loss: Here, we vary the choice of the objective function for Bbox regression to validate the importance of our proposed BDL. Specifically, we first preserve only the objective function for the classification (denoted as $\mathcal{L}_{\text{cls}}$ in Table IV) to see the importance of attacking the Bbox regression. Then, we set another competitor, i.e., the CBL loss [33]. The results are reported in Table IV, where two conclusions can be established. First, attacking only the classification branch shows more inferior threats than taking both Bbox regression into consideration. Meanwhile, the introduction of both BDL and CBL can sacrifice the ℓ_2-norm, resulting in poorer visual quality than using $\mathcal{L}_{\text{cls}}$ only. However, the improvements of the attack strengths are significant compared with the decrease in the visual quality. Thus, introducing the attack on Bbox regression poses more threats than attacking the classification only. Second, compared with CBL, we reach the significant threats with the competitive improvements of the visual quality. We can see that the ℓ_2-norms of CBL are larger than our TPA, which echoes what we discussed in Section IV-C, i.e., the problem of gradient inundation in CBL [33] may risk the stagnation of optimization, leading to the decrease in the attack threats. Besides quantitative results, we visualize the effects of our BDL in Fig. 8. Compared with $\mathcal{L}_{\text{cls}}$ only, the addition of $\mathcal{L}_{\text{BDL}}$ can result in more intense destruction, which can be reflected in the feature maps, where the attention areas are significantly interfered. By contrast, using only $\mathcal{L}_{\text{cls}}$ could not cause significant attacks on intermediate representations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:42:26 UTC from IEEE Xplore. Restrictions apply.

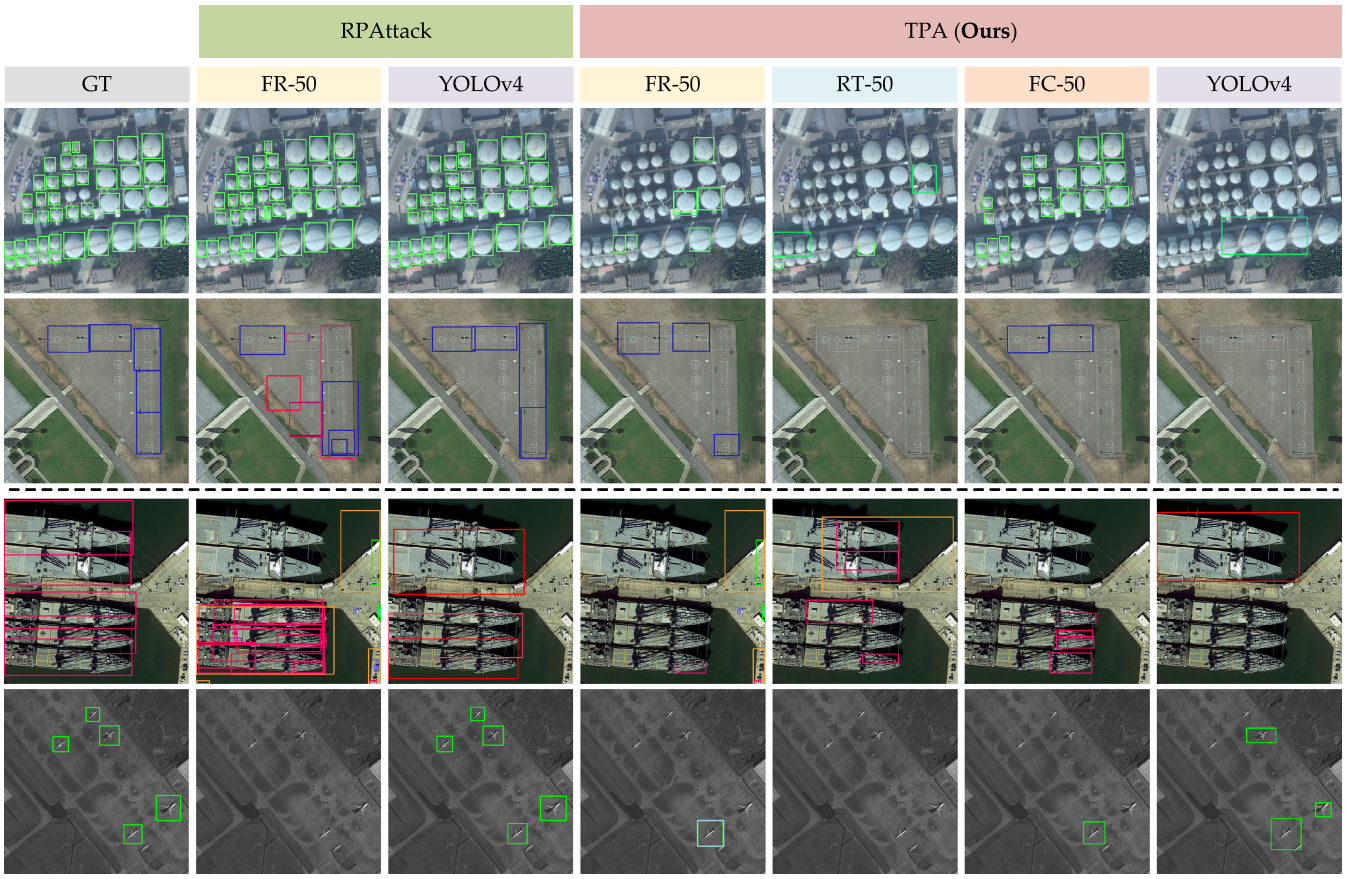


Fig. 7. **Qualitative results with respect to the effect of the BDL.** Here, GT represents ground truth. Samples extracted from DIOR-A and DOTA-A are separated by the dotted line, where the upper samples belong to DIOR-A and the remaining belong to DOTA-A.

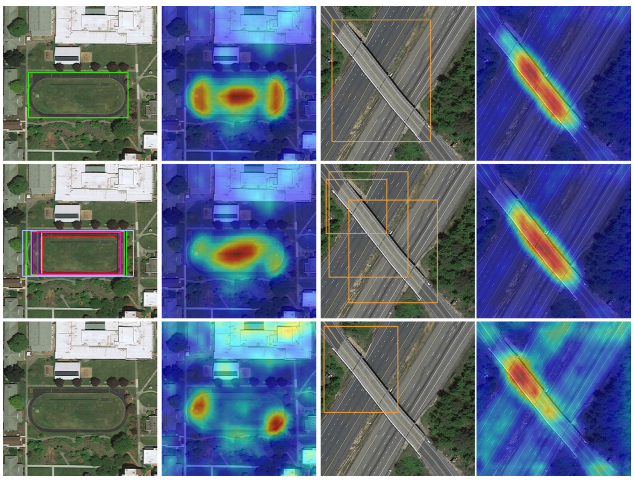


Fig. 8. **Visualizations of the detected results (the singular columns) and the corresponding features (the even columns).** Here, the results on clean images are shown in the first row, the results on adversarial examples generated via $\mathcal{L}_{\text{cls}}$ only are exhibited in the second row, and the others are the results on adversarial examples generated via $\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{BDL}}$.

TABLE V
ABLATION RESULTS REGARDING THE GRID SEGMENTATION SCHEME

Detector	Method	mAP (%)	Recall (%)	ℓ_0 Norm	ℓ_2 Norm
FR-50	U(2)	48.50	56.90	0.065	5.180
	U(3)	49.40	57.50	**0.057**	**4.980**
	SA(1, 2, 3)	47.80	55.80	0.059	4.990
	SA(2, 3, 4)	**46.30**	**55.20**	0.064	5.210
FC-50	U(2)	48.20	63.00	0.065	5.800
	U(3)	49.10	63.50	**0.057**	**5.510**
	SA(1, 2, 3)	47.00	61.50	0.059	5.540
	SA(2, 3, 4)	**45.20**	**59.30**	0.064	5.840
RT-50	U(2)	40.30	58.20	0.065	5.290
	U(3)	41.30	59.90	**0.057**	**5.140**
	SA(1, 2, 3)	39.60	58.00	0.059	5.170
	SA(2, 3, 4)	**38.20**	**55.10**	0.064	5.380

3) Ablation Study on Grid Segmentation Scheme: Since grid segmentation is the first step in the patch selection scheme, there is no study regarding the influence of different schemes on final results. To this light, we propose to provide a preliminary experimental exploration in this section. Recall that we provided two options for the grid segmentation in Section IV-B, i.e., the universal scheme and the scale-adaptive scheme. In this part, we explore the influence on these schemes. Specifically, we set different parameters in U(n) and SA(n_1, n_2, n_3). The results are shown in Table V. Generally speaking, the choice of these schemes does not seem to play a key role in the final results. When we take a closer look at these results, we can find that with the same

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:42:26 UTC from IEEE Xplore. Restrictions apply.

visual effect, the scale-adaptive scheme shows more aggressive ability than the universal scheme, while leaving the visual quality get sacrificed slightly. For instance, SA(1, 2, 3) and U(3) achieve almost the same visual effect, but the attack strength of SA(1, 2, 3) is better than U(3). The same case can be found in the comparison between U(2) and SA(2, 3, 4), where SA(2, 3, 4) exhibits more threats than U(2). Thus, considering the attack strength and visual effect comprehensively, we choose SA(1, 2, 3) as our final choice.

VI. Conclusion

In this article, we paid attention to PAs on object detection in O-RSIs and proposed a threatening PA without the scarification of the visual quality, dubbed TPA. Specifically, to address the problem of inconsistency between the local and global landscapes in existing patch selection schemes, we proposed to leverage the FOD of the objective function before and after masking to select the subpatches to be attacked. Furthermore, considering the problem of gradient inundation when applying existing CBL to PAs directly, we designed an IoU-based objective function specific for PAs, dubbed BDL, which pushes the detected Bboxes far from the initial ones until there are no overlaps between them. Compared with the advanced competitor, the extensive evaluations have witnessed the remarkable effectiveness of our TPA. Moreover, we also replace the key factors of our TPA to see their significant influence in the final results and further reveal the effectiveness of them. We hope this first attempt can arouse the research interest in further works regarding PAs on object detection in O-RSIs.

For future works, whether more information in the detection model could be used to design a more threatening loss function? Can the gradient information be used to reduce the attack regions and maintain the attack strength? These questions will be the research topics worth further exploration.

References

[1] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.

[2] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. A. Alemi, "Inception-v4, inception-ResNet and the impact of residual connections on learning," in *Proc. AAAI Conf. Artif. Intell.*, 2017, pp. 1–7.

[3] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 1–9.

[4] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9627–9636.

[5] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[6] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.

[7] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[8] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[9] G. Cheng, P. Lai, D. Gao, and J. Han, "Class attention network for image recognition," *Sci. China Inf. Sci.*, vol. 66, no. 3, Mar. 2023, Art. no. 132105.

[10] G. Cheng et al., "Anchor-free oriented proposal generator for object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625411.

[11] G. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[12] Q. He, X. Sun, Z. Yan, B. Li, and K. Fu, "Multi-object tracking in satellite videos with graph-based multitask modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5619513.

[13] C. Li, G. Cheng, G. Wang, P. Zhou, and J. Han, "Instance-aware distillation for efficient object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602011.

[14] Y. Yao et al., "On improving bounding box representations for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600111.

[15] Z. Wu, W. Zhu, J. Chanussot, Y. Xu, and S. Osher, "Hyperspectral anomaly detection via global and local joint modeling of background," *IEEE Trans. Signal Process.*, vol. 67, no. 14, pp. 3858–3869, Jul. 2019.

[16] G. Cheng, C. Lang, M. Wu, X. Xie, X. Yao, and J. Han, "Feature enhancement network for object detection in optical remote sensing images," *J. Remote Sens.*, vol. 2021, pp. 1–14, Jan. 2021.

[17] Z. Wu et al., "Scheduling-guided automatic processing of massive hyperspectral image classification on cloud computing architectures," *IEEE Trans. Cybern.*, vol. 51, no. 7, pp. 3588–3601, Jul. 2021.

[18] Z. Wu, J. Sun, Y. Zhang, Z. Wei, and J. Chanussot, "Recent developments in parallel and distributed computing for remotely sensed big data processing," *Proc. IEEE*, vol. 109, no. 8, pp. 1282–1305, Aug. 2021.

[19] Q. He, X. Sun, W. Diao, Z. Yan, D. Yin, and K. Fu, "Transformer-induced graph reasoning for multimodal semantic segmentation in remote sensing," *ISPRS J. Photogramm. Remote Sens.*, vol. 193, pp. 90–103, Nov. 2022.

[20] Q. He, X. Sun, W. Diao, Z. Yan, F. Yao, and K. Fu, "Multimodal remote sensing image segmentation with intuition-inspired hypergraph modeling," *IEEE Trans. Image Process.*, vol. 32, pp. 1474–1487, 2023.

[21] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.

[22] C. Szegedy et al., "Intriguing properties of neural networks," 2013, *arXiv:1312.6199*.

[23] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2017, pp. 39–57.

[24] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–23.

[25] S.-M. Moosavi-Dezfooli, A. Fawzi, and P. Frossard, "DeepFool: A simple and accurate method to fool deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2574–2582.

[26] F. Tramèr, A. Kurakin, N. Papernot, I. Goodfellow, D. Boneh, and P. McDaniel, "Ensemble adversarial training: Attacks and defenses," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–20.

[27] G. Cheng, X. Sun, K. Li, L. Guo, and J. Han, "Perturbation-seeking generative adversarial networks: A defense framework for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605111.

[28] Y. Xu, T. Bai, W. Yu, S. Chang, P. M. Atkinson, and P. Ghamisi, "AI security for geoscience and remote sensing: Challenges and future trends," 2022, *arXiv:2212.09360*.

[29] W. Czaja, N. Fendley, M. Pekala, C. Ratto, and I.-J. Wang, "Adversarial examples in remote sensing," in *Proc. 26th ACM SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, Nov. 2018, pp. 408–411.

[30] Y. Xu, B. Du, and L. Zhang, "Assessing the threat of adversarial examples on deep neural networks for remote sensing scene classification: Attacks and defenses," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1604–1617, Feb. 2021.

[31] C. Xie, J. Wang, Z. Zhang, Y. Zhou, L. Xie, and A. Yuille, "Adversarial examples for semantic segmentation and object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1369–1378.

[32] S.-T. Chen, C. Cornelius, J. Martin, and D. H. P. Chau, "ShapeShifter: Robust physical adversarial attack on faster R-CNN object detector," in *Proc. ECML PKDD*, 2019, pp. 52–68.

[33] Y. Li, D. Tian, M.-C. Chang, X. Bian, and S. Lyu, "Robust adversarial perturbation on deep proposal-based models," 2018, *arXiv:1809.05962*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:42:26 UTC from IEEE Xplore. Restrictions apply.

[34] H. Zhang, W. Zhou, and H. Li, "Contextual adversarial attacks for object detection," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2020, pp. 1–6.

[35] O. Mohamad Nezami, A. Chaturvedi, M. Dras, and U. Garain, "Pick-object-attack: Type-specific adversarial attack for object detection," *Comput. Vis. Image Understand.*, vol. 211, Oct. 2021, Art. no. 103257.

[36] S. Wu, T. Dai, and S.-T. Xia, "DPAttack: Diffused patch attacks against universal object detection," 2020, *arXiv:2010.11679*.

[37] Y. Zhao, H. Yan, and X. Wei, "Object hider: Adversarial patch attack against object detectors," 2020, *arXiv:2010.14974*.

[38] H. Huang, Y. Wang, Z. Chen, Z. Tang, W. Zhang, and K.-K. Ma, "Rpattack: Refined patch attack on general object detectors," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2021, pp. 1–6.

[39] T. B. Brown, D. Mané, A. Roy, M. Abadi, and J. Gilmer, "Adversarial patch," 2017, *arXiv:1712.09665*.

[40] X. Liu, H. Yang, Z. Liu, L. Song, H. Li, and Y. Chen, "DPatch: An adversarial patch attack on object detectors," 2018, *arXiv:1806.02299*.

[41] M. Lee and Z. Kolter, "On physical adversarial patches for object detection," 2019, *arXiv:1906.11897*.

[42] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2117–2125.

[43] Y. Dong, T. Pang, H. Su, and J. Zhu, "Evading defenses to transferable adversarial examples by translation-invariant attacks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4312–4321.

[44] C. Xie et al., "Improving transferability of adversarial examples with input diversity," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2730–2739.

[45] J. Lin, C. Song, K. He, L. Wang, and J. E. Hopcroft, "Nesterov accelerated gradient and scale invariance for adversarial attacks," in *Proc. Int. Conf. Learn. Represent.*, 2020, pp. 1–12.

[46] Y. Dong et al., "Boosting adversarial attacks with momentum," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9185–9193.

[47] W. Brendel, J. Rauber, and M. Bethge, "Decision-based adversarial attacks: Reliable attacks against black-box machine learning models," 2017, *arXiv:1712.04248*.

[48] X. Sun, G. Cheng, L. Pei, and J. Han, "Query-efficient decision-based attack via sampling distribution reshaping," *Pattern Recognit.*, vol. 129, Sep. 2022, Art. no. 108728.

[49] X. Sun, G. Cheng, H. Li, L. Pei, and J. Han, "Exploring effective data for surrogate training towards black-box attack," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15355–15364.

[50] M. Zhou, J. Wu, Y. Liu, S. Liu, and C. Zhu, "DaST: Data-free substitute training for adversarial attacks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 234–243.

[51] S. H. Silva and P. Najafirad, "Opportunities and challenges in deep learning adversarial robustness: A survey," 2020, *arXiv:2007.00753*.

[52] A. Athalye, L. Engstrom, A. Ilyas, and K. Kwok, "Synthesizing robust adversarial examples," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 284–293.

[53] A. Liu et al., "Perceptual-sensitive GAN for generating adversarial patches," in *Proc. AAAI Conf. Artif. Intell.*, 2019, vol. 33, no. 1, pp. 1028–1035.

[54] J. Lian, S. Mei, S. Zhang, and M. Ma, "Benchmarking adversarial patch against aerial detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5634616.

[55] R. den Hollander et al., "Adversarial patch camouflage against aerial detection," *Proc. SPIE*, vol. 11543, pp. 77–86, Sep. 2020.

[56] M. Lu, Q. Li, L. Chen, and H. Li, "Scale-adaptive adversarial patch attack for remote sensing image aircraft detection," *Remote Sens.*, vol. 13, no. 20, p. 4078, Oct. 2021.

[57] S. Thys, W. V. Ranst, and T. Goedemé, "Fooling automated surveillance cameras: Adversarial patches to attack person detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2019, pp. 49–55.

[58] J. Wang, A. Liu, Z. Yin, S. Liu, S. Tang, and X. Liu, "Dual attention suppression attack: Generate adversarial camouflage in physical world," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8565–8574.

[59] A. Kurakin, I. J. Goodfellow, and S. Bengio, "Adversarial examples in the physical world," in *Proc. Artif. Intell. Safety Secur.*, 2018, pp. 99–112.

[60] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.



Xuxiang Sun received the B.E. degree from Nanchang University, Nanchang, China, in 2019. He is currently pursuing the Ph.D. degree with Northwestern Polytechnical University, Xi'an, China.

His research interests include computer vision and image processing, especially on generative models and adversarial robustness of deep learning.



Gong Cheng (Member, IEEE) received the B.S. degree from Xidian University, Xi'an, China, in 2007, and the M.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, in 2010 and 2013, respectively.

He is a Professor with Northwestern Polytechnical University. His main research interests are computer vision, pattern recognition, and remote sensing image understanding.

Dr. Cheng is an Associate Editor for *IEEE Geoscience and Remote Sensing Magazine*.



Lei Pei received the B.E. degree from Northwestern Polytechnical University, Xi'an, China, in 2020, where she is pursuing the master's degree, under the supervision of Prof. Gong Cheng.

Her main research interests are object detection, remote sensing image understanding, and adversarial attacks.



Hongda Li received the B.E. degree from Northwestern Polytechnical University, Xi'an, China, in 2021, where he is pursuing the master's degree, under the supervision of Prof. Gong Cheng.

His main research interests are object detection, remote sensing image understanding, and adversarial attacks.



Junwei Han (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees in pattern recognition and intelligent systems from Northwestern Polytechnical University, Xi'an, China, in 1999, 2001, and 2003, respectively.

He was a Research Fellow with Nanyang Technological University, Singapore, The Chinese University of Hong Kong, Hong Kong, the Dublin City University, Dublin, Ireland, and the University of Dundee, Dundee, U.K., from 2003 to 2010. He is a Professor with Northwestern Polytechnical University. His research interests include computer vision and brain imaging analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:42:26 UTC from IEEE Xplore. Restrictions apply.